

Lucio Find: Faster Legal Document Discovery & Assistance

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1. Problem Statement

The manual search process everywhere works on a simple fundamental concept: “Look for the keywords mentioned, and present all the files that have even one of the keywords mentioned in them.” For lawyers, there’s not one document or one single case file. And it’s not necessary that they do remember the right file name.

It could be natural language query, not a filename, a concept. Something like "share purchase agreement with ABC Corp indemnity clause" or "NDA template we use for Series B deals."

1. What problem is the lawyer facing?

- Lawyers’ *speeds slow down even before the official work starts*. Knowing what they need, but being not able to locate it quickly, is a loss not just to the lawyer but to the firm too.
- Most firms work with various technologies to manage a single deal. Now, the *docs for each are multi-threaded to various platform- local folder, previous Lucio session or forgotten email thread*, increasing the search time immensely.
- As per LinkedIn, just *15 minutes to search daily* for documents, attorneys add up to *60 hours annually, roughly 7.5 full working days lost/lawyer/year*.

2. Why is this problem important?

- *Expensive lost time, with an invisible cost.*
- *Avg lawyer works for 48 hrs., but bills for only 36 hr.* The 12 hrs. gap does include the time wasted on searching on doc.
- An overhead burnout session. Lucio’s official data states, organizations lose approximately *9.2% of annual revenue to contract inefficiency alone*.

3. What are lawyers trying to do after finding the document?

- *Finding the doc is an entry point*. After locating the right document, the actual work starts, which could be review a clause, compare against a current draft.
- The *legal profession involves many repetitive tasks* such as document review, contract analysis, and e-discovery, these tasks, while necessary, take up valuable time that could be *better spent advising clients, preparing for trial, or strategic planning*.
- *Actual job is the judgement, the drafting, the advice which begins after they have the right raw material aka “the document”*.

4. Why is normal keyword or folder search not enough?

There are three valid reasons for the above:

- Legal works requires concepts not keyword:** In fast-paced environments like hearings, drafting sessions, or consultations, you *may only remember a concept or situation, not the legalese*. You might know the issue *at hand, but without the precise article number or legal phrase*, traditional search tools leave you chasing *dead ends or buried in irrelevant results*.
- Keywords have no context:** *Matching words than the meaning is irrelevant* because the *context is missing*.

For an instance, you search a word “indemnity negotiated”, every file with that word would pop up. You’d still have to open each and read through them.

- iii. **No traditional search knows about the lawyer’s history or intent:** *The unique context generated by the law firm itself is the source of the value.*

2. Users and Use Cases

User1: M&A Associates

- i. **Who are they:** A corporate professional, typically in finance or law who manages the transaction process, and preparing investment memos.
- ii. **What they search for:** "indemnity cap SPA German counterparty 18 months" or "cross-border acquisition disclosure letter template." The deal, not the filename.
- iii. **The problem:** *Average of 11.2 hours per week is spent while dealing with document-related challenges, document mismanagement costs ~\$9,071 per lawyer annually, reducing firm productivity to nearly 10%.*

User2: General Counsel

- i. **Who they are:** A solo GC at a 200-person SaaS company, handling every legal matter themselves, with a deadline approaching by EOD.
- ii. **What they search for:** "previous MSA with similar vendor what did we agree on data processing?" They need their own company's position, not generic templates.
- iii. **The problem:** *55% of in-house counsel do not have a single, standardized contract management platform. 43% of in-house legal professionals say contracts-related tasks represent at least half of their daily work, and 71% manage a high volume of contracts at once.*

User3: Senior Associate doing Litigation

- i. **Who they are:** A senior associate leading a due diligence workstream having 300 target company documents & need to find every contract that contains a change of control clause before a 48-hour deadline.
- ii. **What they search for:** "change of control provisions" across 300 contracts which is a conceptual search that keyword tools handle poorly, because the clause might say anything, with variations depending on who drafted it.
- iii. **The problems:** *When attorneys spend just 15 minutes daily searching for documents, adding up to 60 hours annually, costing firms ~\$18,000/lawyer. Search problem is compressed into 48 hours, the stakes of missing a clause are deal-breaking, not just inefficient.*

3. Proposed Solution

Lucio Find, a universal search layer build into Lucio. Sitting one step before Lucio Assistant, its job is to *find files, based on the memory search and then bringing relevant files with a hint of snippet that why, is it relevant.*

1. Where to search?

- First, as a persistent search bar at the top of the Lucio Sidebar, always visible
- Second, default empty state when a lawyer opens a new Lucio Assistant session, opening with a search bar that says “find a document to work with”.

2. What the user types?

- *Anything natural, a concept, a deal name, a clause type or a combination.* For example, “*change of control provisions Nexus acquisition*”
- Find, will interpret the query semantically, understanding the “change of control provision” and “acquisition clause” as similar concept.

3. What sources they can search across?

By default, *all sources will be searched.* But *privacy concerns*, so we allow the firm to on or off the toggles for each source.

The major sources are:

- Lucio Sessions:** The memory layer where the search will begin first. Every doc the previously opened, downloaded or queried will be searched. Highest context, with who worked on it last, what was done etc.
- Lucio Repository:** Firm-level documents stored centrally in Lucio.
- Connected DMS:** Documents in the firm's document management system, accessed via API. Requires a one-time connection by the firm's admin.
- Local Matter Folders:** Files on the lawyer's own machine or mapped network drives, indexed by Lucio Studio.
- Expand further to Web and legal databases, Email attachments etc.**

4. What the search result shows?

Each search is a card with five things:

- Document Name
- Source Badge, for each access navigation.
- Relevance Reason, a readable explanation by AI. For instance, "Firm template for cross-border M&A indemnity clauses. Includes German law fallback language."
- Memory Signal. A badge for the user to understand that it had been worked on. It would say something like: "*Redlined in session with Sarah J.*" This is the institutional memory layer surfacing contextual history inline.**
- Two actions, Open and Send to Assistant.

5. How the user opens a document?

- Click **Open**, a side panel slides open showing the document content, with the matched clause or section highlighted. The lawyer can read it, confirm it is the right document, and decide what to do next, without navigating away from Find.
- If the document lives in an external DMS or local folder and cannot be rendered directly, Lucio shows a preview of the relevant extracted section and provides a direct deep link to open the full document in the native system.

6. How to send the document to the Assistant?

- Each card will have a **send to Assistant button**. Click and load the doc directly into the Assistant. We can also have, *multiple checkboxes for multiple documents*, select which one to be opened in the Assistant box.

7. What should the assistant should allow them to do next?

After doc loaded, a brief AI generated message summarizing the document from the memory and metadata is displayed, asking what should be our next steps?

The next steps could be summarizing the doc, comparing visions, extracting etc.

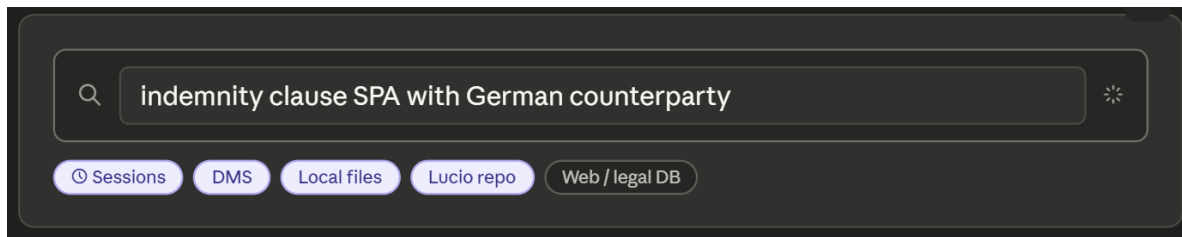


Figure 1: Search Bar

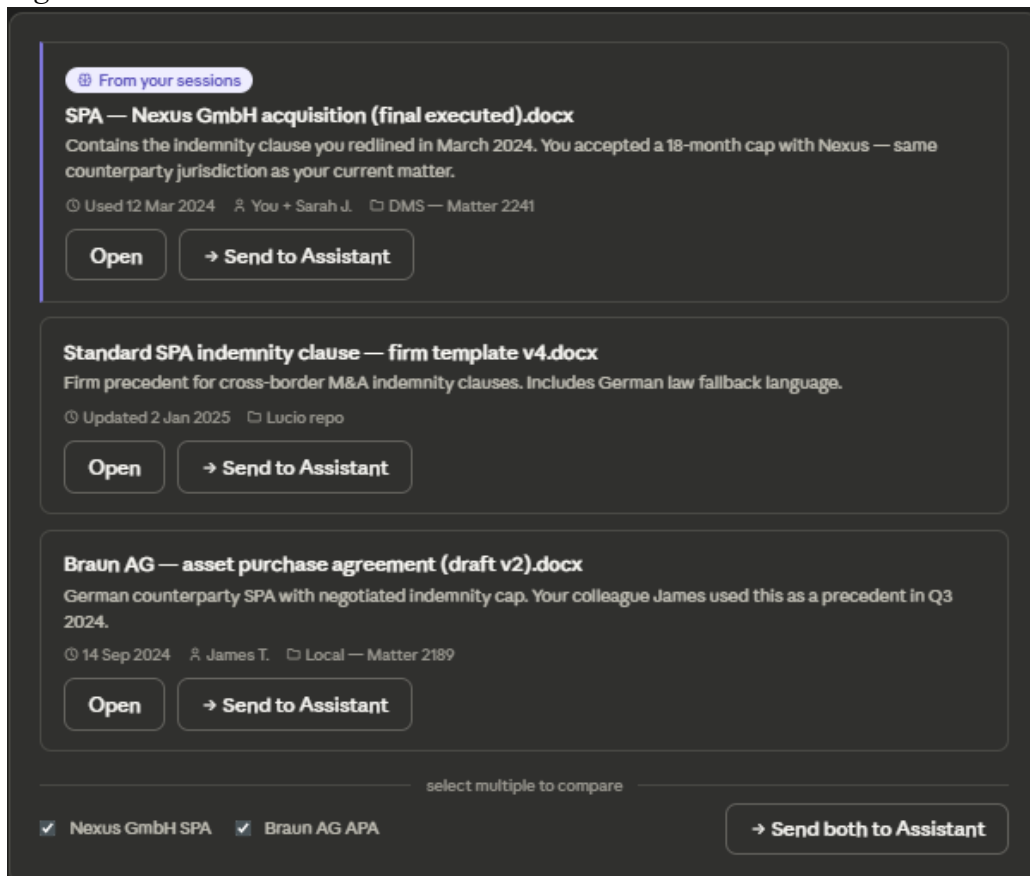


Figure 2: Search Result

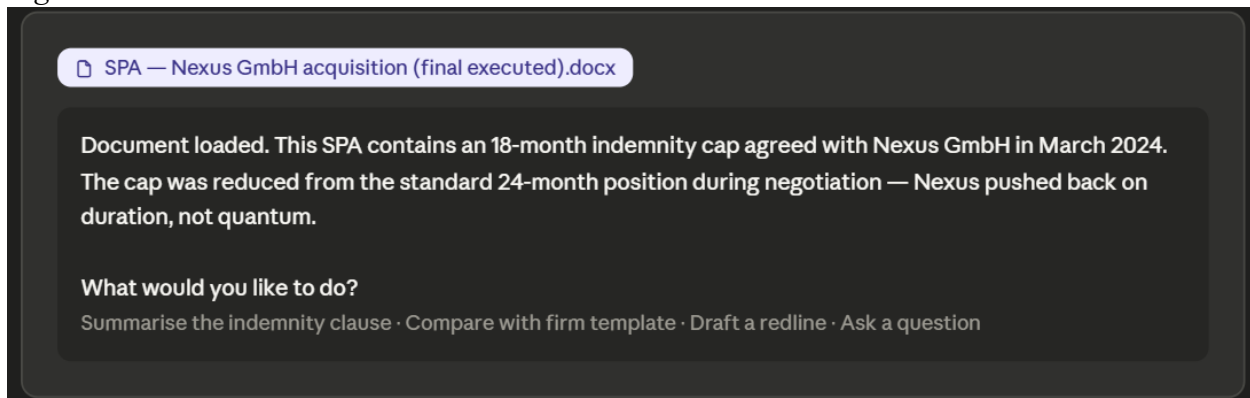


Figure 3: Document loaded on Lucio Assistant

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Lawyer opens new Lucio session
↓
Lucio Find search bar is the default screen
↓
Lawyer types: "indemnity SPA German counterparty"
↓
Find searches all 5 sources simultaneously
↓
Results appear — ranked, with relevance reasons
Memory badge highlights a document from March 2024
↓
Lawyer clicks Open → previews the indemnity clause in side panel
Confirms this is the right document
↓
Lawyer clicks Send to Assistant
↓
Assistant opens warm — summarises the document and negotiation history
↓
Lawyer asks: "Draft a redline of the indemnity clause
              based on our current firm position"
↓
Assistant drafts. Lawyer reviews and edits.
Work is done.

```

Figure: Simple User Flow

4. How could the experience be trustworthy?

- i. **Source Transparency:** Every result will show its provenance completely, with a full file path, the date of last modification, and whether the version is final executed version or draft.
- ii. **Relevance Reason:** The reason would be verifiable. Like click on the exempt, and redirect to the exact passage in the doc, highlighted.
- iii. **Memory Signal:** The history should be verified. And if it isn't then, then there should be a recalled version saying I know we worked on it but let me check once again.
- iv. **Staleness detection:** A document from 2024 might not be what the firm is positioning now in 2026. A staleness check with a flag stating we have newer versions of this document.

5. Prioritization: 2 Weeks Implementation Plan

Week 1 Plan

Goal: Search → ranked results → send to Assistant. End-to-end flow working by Friday.

Day 1 — Entry Point

- Ship search bar as default screen on new Assistant sessions + sidebar
- Backend starts vector indexing on Lucio Repository

Day 2 — Index Sources

- Repository fully indexed into vector store
- Session history lookup built — date + last action per document per lawyer

Day 3 — Results + Relevance

- Search bar wired to semantic engine
- Result cards ship: name, source badge, date, relevance reason with citation link
- Memory badge surfaces on results with session history match

Day 4 — Preview + Send

- Side panel preview: document opens with matched passage highlighted

- Send to Assistant: one click, document loads, warm opening message
- Not Relevant button added to each card

Day 5 — QA + Internal Test

- Test with real firm documents
- QA focus: citation accuracy, memory badge correctness, Send reliability
- Fix breaking issues & try Beta testing with 1–2 lawyers for same-day feedback

Week 2 Plan

Goal: Stabilise v1 from real feedback. Add multi-document send, ranking refinement, and begin local file indexing.

Day 6 — Act on Week 1 Feedback

- Review feedback & fix relevance reason inaccuracies flagged by users
- Improve warm Assistant message copy based on real usage

Day 7 — Ranking Refinement

- Incorporate Not Relevant signals from Day 5 into ranking model
- Boost results by recency + frequency of use per lawyer

Day 8 — Multi-Document Send

- Add checkboxes to result cards
- Build batch Send to Assistant — all selected docs load into one session
- Add context window guard: warn lawyer if selection exceeds limit

Day 9 — Local File Indexing (Lucio Studio)

- Begin indexing local matter folders via Lucio Studio
- QA: confirm no latency issues from local indexing running in background

Day 10 — Hardening + Launch Prep

- Confirm trust signals are accurate: citations, memory badges, version dates
- Internal demo to wider team

6. Risks and Edge Cases

- **Search Quality Risk:** Semantics Mismatch, Hallucination of Relevant Reason, Low Result Volume
- **Memory Layer Risk:** False Memory badge, or incorrect Stale Signal
- **Send to Assistant Risk:** Context window overflow (more files than the assistant can actually work on), Docs can't be rendered

7. Success Metric

North Star Metric: Avg time from opening Lucio to the user asking their first Assistant query.

- If the lawyers are finding the right document faster, this drops.
- Target: Reduce this current baseline by 50% in 30 days of launch

Search Quality:

Metric	Target	Importance
Search to send rate	>60% of searches result in a send to assistant	Measures if the result is good enough to work on
Not relevant rate	<20% of results dismissed	High dismissal=poor relevance
Zero result rate	<10% of searches	Frequent dead ends=Trust kill
Avg result before send	<3	Scrolling time and order should be less and not be deep.

Memory Layer:

Metric	Target	Importance
Memory badge click rate	>30% of badged results clicked	Signals lawyers find session history useful
Memory-Assisted Sends	>25% of all sends involve a memory badged result	Measures memory layer driving real decisions
Memory Accuracy Rate	>90% correct on spot checks	One bad memory signal erodes trust fast

Trust Signals:

Metric	Target	Importance
Citation link click rate	>40% of results opened	Lawyers verifying relevance reasons means healthy behaviour
Relevance Reason Accuracy	>85% on manual QA sample	AI characterization must match actual document content
Staleness flag acknowledgement	>70% of flagged results acknowledged before send	Lawyers reading warning, not ignoring them

Assistant Quality Post-Find:

Metric	Target	Importance
First query success	>75% of post-Find sessions reach a useful output	Document sent was the right one
Multi doc send rate	>20% of sends within 45 days of launch	Signals lawyers discovering comparison workflows
Session length post find	Longer than sessions without find	Right document in context means deeper, more productive work

Warning Signs:

- Not Relevant Rate above 30%
- Zero Result rate above 15%
- Memory badge accuracy drops to 85% or low
- Find-first session rate flat after 2 weeks

8. AI Usage declaration & References used

Tool and Reference	Use
GPT	Structure the note, Formatting
Claude	In-Depth research - legal AI market, What's new, relevancy rate. Competitors, and their work User Flow diagrams (given the idea for the pictorial description) Prototype structure (an ideal prompt was given to get the picture) Prioritization Plan Implementation Success metric: what, and how
LinkedIn	Lawyer Problems, Document Search Dependency
Lucio AI	Metric Data
Viewpointanalysis.com	Articles related to Keyword Search Reliability
Businesswire.com	Articles related to Users and Use cases
Clutch.com	Articles related to Working hours and billings for lawyers
Luminance	About their new memory implementation in the field of negotiations, making it world's first implement of its kind

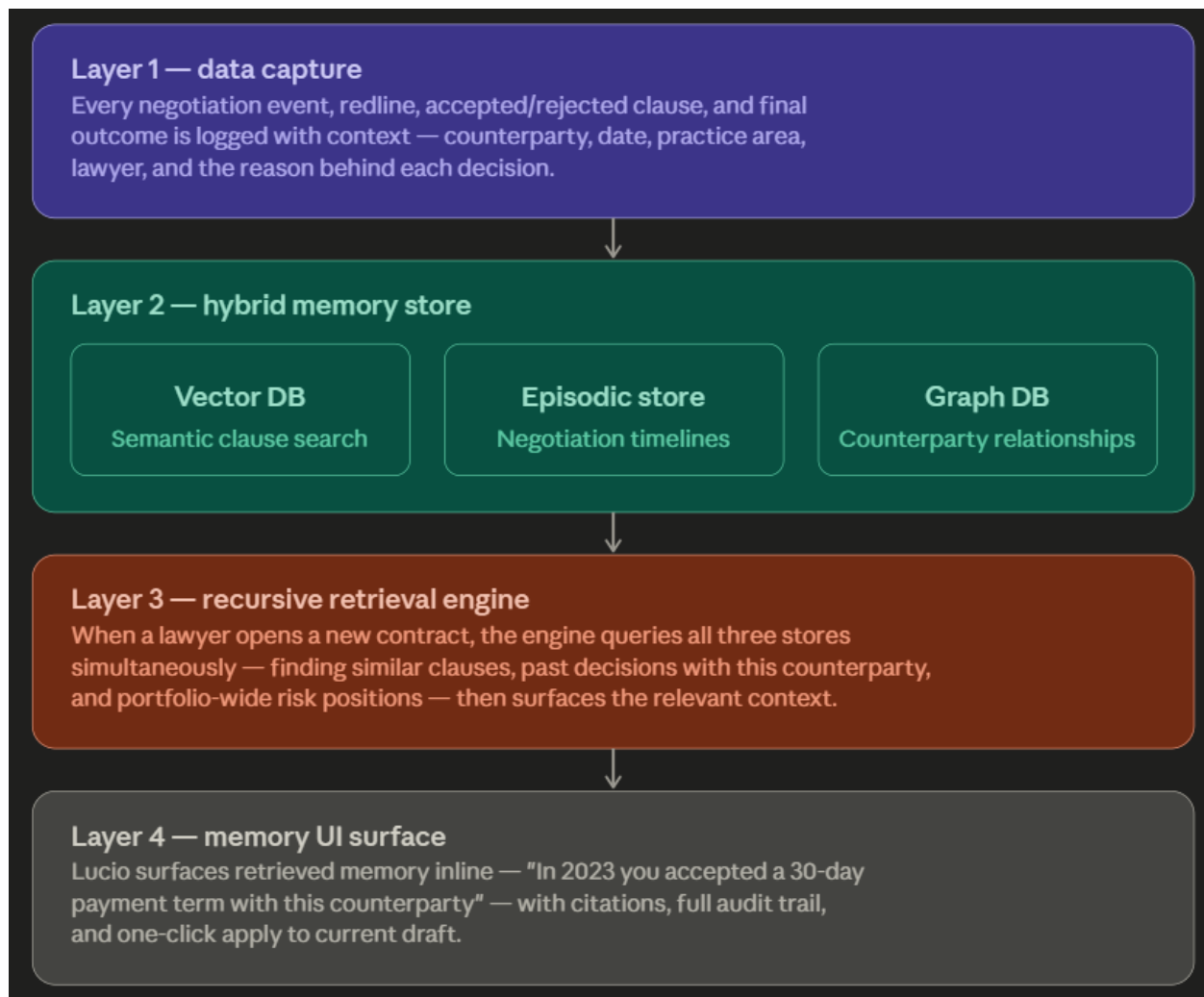


Figure: Implementation of Memory Layer within the Lucio AI